

Coordination Regimes of Synchronised Agents: A Self-Contained Variational Calculus on the S-Entropy Manifold, with External Validation

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July 2, 2026

Abstract

We develop a self-contained variational calculus for the coordination of synchronised agents. An agent is modelled as an overdamped (Onsager–Machlup) dynamical system on a five-dimensional manifold $\mathcal{M} = [0, 1] \times [0, 2\pi^2] \times [0, 1]^3$ whose coordinates are an internal Kuramoto order parameter R , a phase variance σ^2 , and three *S-entropy* coordinates (S_k, S_t, S_e). Crucially, and in contrast to a preliminary version of this work, every construct used here—the S-entropy coordinates, the partition potential, and the governing Lagrangian—is *derived within the paper* from three stated axioms (bounded phase space, categorical exclusion, finite resolution); the manuscript depends on no external or unpublished result. We prove: (i) a triple-equivalence identity establishing the S-entropy coordinates from maximum-entropy counting; (ii) the explicit Euler–Lagrange equations of the agent action, given in full rather than asserted; (iii) the Kuramoto synchronisation bifurcation, with the *corrected* critical coupling $K_c = 2/[\pi g(0)]$ (for a Gaussian frequency law, $K_c = \sqrt{2\pi} \sigma_\omega (2/\pi) \approx 1.596 \sigma_\omega$), replacing an erroneous coefficient in the preliminary version; (iv) a formal, analogy-free definition of *partition extinction* and an observable-commutation theorem that separates functional from ontological indistinguishability; and (v) a coordination calculus for ensembles. We are explicit that the five coordination *bands* on R are operational classification cut-offs, not phase transitions: the only genuine transition in the

model is the Kuramoto bifurcation. Rather than the self-referential numerical checks of the preliminary version, we report *nine external experiments* against public data (atomic shell structure, the Kuramoto onset by direct simulation, the Wiedemann–Franz constant across eleven metals, sleep-EEG order parameters from PhysioNet, enzyme kinetics, antidepressant meta-analysis, superconductor transport, and representation-disjoint classifiers on a benchmark dataset). Two of these external tests forced corrections to the theory and are reported as such. We are deliberately narrow about applicability: the S-entropy coordinates measure the *shape* of an agent’s state distribution, not the content or ecological salience of its information; the model is stochastic (Langevin) but single-scale and on a fixed manifold, so it does not represent the multiscale fluctuation and metastability of real biological collectives; and we exhibit a concrete interpretable Lagrangian and a real-data ensemble instance rather than claim to capture any such collective in full. A Scope and Limitations section delimits the defensible claims.

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3	Single-agent dynamics: potential, action, and explicit Euler–Lagrange equations	7	This paper develops a variational calculus for the coordination of synchronised agents and tests it against external data. A preliminary version of this work was written as one entry in a larger programme and, in consequence, relied on constructions (an “S-entropy” coordinate system, a “cell-truth” receiver calculus, a partition Lagrangian) that were defined in companion manuscripts rather than in the paper itself. Reviewers correctly identified this as a defect: a reader cannot verify a theory whose primitives live elsewhere. The present version is rewritten to be <i>mathematically self-contained</i> . Every object is constructed here from three axioms; every theorem is proved from those axioms and standard mathematics; and no claim depends on an unpublished result.	
3.1	The partition potential	7	A second defect of the preliminary version was its empirical section, which reported fifty numerical “experiments” agreeing with theory “to machine precision.” As reviewers noted, those checks evaluated closed-form predictions against the same closed forms; agreement to 10^{-16} reflects floating-point rounding of identities, not validation. We replace that section with nine <i>external</i> experiments (Section 7) that compare the theory to data it did not generate. Two of the nine contradicted the preliminary theory and forced corrections, which we make explicit. We regard the presence of genuine corrections as evidence that the remaining con-	
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firmations are meaningful.

1.2 The coordination phenomenon

Agents with markedly different internal organisation routinely converge on common outcomes: drivers avoid collisions, independent laboratories replicate a finding through different instruments, members of an ensemble lock to a shared tempo. The classical account grounds such coordination in common knowledge or focal points [14, 19, 2], an idealisation that is unreachable for agents with mutually inaccessible representations [7]. We pursue a different account: coordination is synchronisation of a collective order parameter, and the deepest coordinated state is one in which inter-agent distinctions cease to be operative. We make this precise and, where the data permit, test it.

1.3 What the framework claims, and what it does not

A reviewer rightly cautioned that the applicability of such a formalism is easily overstated, particularly for biological collectives such as social-insect colonies and neural systems. We therefore state the reach of the theory narrowly at the outset, and repeat the delimitation as a formal Scope section (Section 8). Three points frame everything below. (i) *Content-blindness*. The S-entropy coordinates are normalised Shannon entropies—measures of the *shape* of an agent’s distribution over its partition states. They do not encode ecological salience or the content of the information a biological agent attends to; two agents with identical S-entropy geometry may be responding to entirely different things (Remark 2.12). The framework models the informational geometry of coordination, not its semantics, and we make no claim to the latter. (ii) *The model is stochastic but single-scale*. The primitive object is a Langevin diffusion (Definition 3.3); the Lagrangian scores its path probabilities, and the deterministic gradient flow is only the zero-noise most-probable path (Remark 3.5). Real colonies and nervous systems are stochastic across *many* scales with fluctuating, sometimes metastable, network structure; we represent noise at a *single* scale and on a *fixed* manifold, and we do not model multiscale fluctuation or a time-varying phase space. (iii)

The empirical status is modest and mixed. We validate against public data (Section 7); several tests confirm within a stated scope, two *contradict* the theory and force corrections, and one biological claim (a sleep-stage ordering) is *withdrawn* on real EEG. We exhibit a concrete, interpretable instance of the single-agent Lagrangian (Remark 3.6) and a real ensemble instance (EXT08), but we do not claim the formalism captures the full dynamics of any biological collective. The defensible contribution is a self-contained variational language for order-parameter coordination, together with an honest account of where it does and does not fit the world.

1.4 Contributions, and what is and is not new

We separate genuinely new contributions from imported standard results, as reviewers requested.

Imported (standard, used as tools): the Kuramoto model and its mean-field bifurcation [13, 21, 1, 6]; the Onsager–Machlup action for overdamped dynamics [17, 15, 10]; the Banach fixed-point theorem [3]; maximum-entropy inference [20, 5]; and the Wiedemann–Franz law and partition-function counting from standard physics.

New here: (1) the derivation of the S-entropy coordinate manifold from three axioms via a triple-equivalence identity (Section 2); (2) the explicit partition potential and the full Euler–Lagrange system of the agent action (Section 3); (3) a precise, analogy-free definition of partition extinction together with an observable-commutation theorem distinguishing functional from ontological indistinguishability (Section 6); (4) the identification of the coordinated/phase-locked regime with partition extinction *as a structural correspondence, not an identity*; and (5) the external-validation programme of Section 7, including the corrected critical coupling.

1.5 Relation to existing theories, and what the Lagrangian buys

To let the reader locate the incremental advance, we state the relation to the three nearest bodies of work explicitly.

Versus Kuramoto synchronisation. The classical Kuramoto model describes phase oscillators and their mean-field bifurcation; it is a model of *phases*. We use it unchanged as the R -sector of our dynamics (Theorems 3.9 and 3.10). What is added is not a new synchronisation mechanism but an *enveloping state space*: the synchronisation variable R is one of five coordinates on $\mathcal{M}$, coupled through a single scalar potential Φ to a phase variance and to three informational (S-entropy) coordinates, so that coordination, variance control, and informational geometry are governed by one variational principle rather than three separate models. Kuramoto alone has no informational coordinates and no notion of partition extinction.

Versus receiver / “cell-truth” accounts of coordination. Accounts that ground coordination in shared representation or common knowledge require agents to converge in *content*; they are unreachable for agents with mutually inaccessible representations. Our conservative claim (Theorem 5.2) is the opposite: the ensemble order parameter depends only on (R_i, ϕ_i) and *not* on internal frameworks, so coordination on an outcome is possible with disjoint representations—demonstrated on real representation-disjoint classifiers in EXT08. This replaces a shared-representation premise with a synchronisation premise.

Versus a physical theory of indistinguishability. Partition extinction is here a *classical, information-theoretic* notion (Definition 6.3): binary categorical indistinguishability under a finite-resolution axiom, with an observable-commutation theorem (Theorem 6.7) separating this *functional* equivalence from the *ontological* indistinguishability of identical quantum particles. We do not import quantum statistics; the physical condensates are tested only as examples (EXT07).

What the Lagrangian buys. The variational formulation is not decoration. It yields three things a bare synchronisation model does not: (a) a *single* object Φ whose stationary points classify all rest states of the five-coordinate system at once, so coherence, variance floor, and informational relaxation are consequences of one minimisation rather than separate postulates; (b) explicit equations of motion by one Euler–

Lagrange computation (Theorem 3.7) with the deterministic flow arising as the zero-noise most-probable path of a genuine stochastic model (Remark 3.5); and (c) a principled cost functional—the Onsager–Machlup action assigns a probability to every path—so that “coordination cost” is defined, not stipulated. Where a traditional model would add each effect by hand, the Lagrangian derives them jointly from Φ ; that is its irreplaceable advantage, and also the reason its scope is bounded by the adequacy of Φ , which we test rather than assume.

Table 1: Location relative to the nearest theories. “New here” marks the incremental content of this paper; imported rows are used unchanged as tools.

Ingredient	Nearest prior work	Status here
Phase synchronisation, $K_c = 2/[\pi g(0)]$	Kuramoto mean-field	imported, R -sector only
Path action for dissipative dynamics	Onsager–Machlup	imported, scores paths on $\mathcal{M}$
Five-coordinate manifold $\mathcal{M}$ with single potential Φ	—	new
S-entropy coordinates from triple equivalence	max-entropy counting	new derivation
Analogy-free partition extinction + commutation	(quantum indistinguishability, by contrast)	new , classical
Coordination without shared representation	common-knowledge accounts	new claim, tested (EXT08)

1.6 A note on the five “regimes”

The preliminary version asserted that the order parameter R separates five regimes at $R \in \{0.3, 0.5, 0.8, 0.95\}$ and that these boundaries are second-order phase transitions. We retract that assertion. The boundary values are *operational classification cut-offs*—convenient labels for bands of a continuous variable—and we show below that they are not phase transi-

tions: the partition potential has no curvature sign-change at them. The only genuine transition in the model is the Kuramoto synchronisation bifurcation at K_c (Section 3). This honest demotion removes a circular argument present in the preliminary version (regimes defined by thresholds, thresholds then “derived” from the regimes) and is consistent with the fact that companion analyses used different numerical cut-offs for the same qualitative bands.

1.7 Organisation

Part I states the axioms and derives the S-entropy manifold, the partition potential, the agent Lagrangian, and the explicit Euler–Lagrange equations. **Part II** develops single-agent dynamics and the Kuramoto bifurcation with the corrected critical coupling. **Part III** develops ensembles and the coordination calculus. **Part IV** gives the analogy-free theory of partition extinction. **Part V** reports the nine external experiments, the Scope and Limitations, and the conclusion.

2 Axioms and the S-entropy manifold

We adopt three axioms. They are deliberately weak; their role is to fix the state space and its counting structure, not to encode the conclusions.

Axiom 2.1 (Bounded phase space). The accessible phase space Ω of an agent has finite measure, $\mu(\Omega) < \infty$.

Axiom 2.2 (Categorical exclusion). At each instant the agent occupies exactly one of a finite set $\mathcal{C}$ of mutually exclusive categorical states; there is no null state.

Axiom 2.3 (Finite resolution). Every observation partitions Ω into finitely many distinguishable cells; states closer than a resolution $\delta > 0$ are indistinguishable.

Remark 2.4. Axioms 2.1 to 2.3 are satisfied by essentially every physical or computational system, which is both their strength (generality) and their weakness (they do not, by themselves, discriminate among systems). We flag this explicitly; the discriminating content of the theory

is in the dynamics built on top of them, and in the external tests of Section 7, not in the axioms.

2.1 Partition coordinates and capacity

Definition 2.5 (Partition coordinates). Under Axioms 2.1 to 2.3 a bounded three-dimensional oscillatory state is labelled by a tuple (n, ℓ, m, s) with $n \in \mathbb{N}$, $0 \leq \ell < n$, $-\ell \leq m \leq \ell$, and $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$, arising from the nested closure conditions of bounded oscillation (radial depth n , angular index ℓ , orientation m , chirality s).

Theorem 2.6 (Shell capacity). *The number of partition states at depth n is $C(n) = 2n^2$.*

Proof. For each $\ell \in \{0, \dots, n-1\}$ there are $2\ell + 1$ values of m and two values of s , so $C(n) = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) = 2n^2$. $\square$

Remark 2.7. We test $C(n) = 2n^2$ against the real periodic table in Section 7 (EXT01). The formula is exact as a level degeneracy; we will be careful there to distinguish it from the question of shell *closure*, which requires the Aufbau filling order and which $2n^2$ alone does not determine.

2.2 The triple equivalence and the S-entropy coordinates

Definition 2.8 (Three entropies). For M independent modes each occupying one of n distinguishable states: the *oscillatory* entropy $S_{\text{osc}} = -k_B \sum_j p_j \ln p_j$ over mode occupations p_j ; the *categorical* entropy $S_{\text{cat}} = k_B \ln W$ over the number W of consistent categorical microstates; and the *partition* entropy $S_{\text{part}} = k_B \ln C(n)^M$.

Theorem 2.9 (Triple equivalence). *At maximum entropy over the $C(n)$ states of depth n , the three entropies of Definition 2.8 coincide:*

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_B M \ln n + \text{const},$$

where the constant absorbs the n -independent term $k_B M \ln 2$.

Proof. By the maximum-entropy principle [20, 5], the distribution over the $C(n) = 2n^2$ states

maximising S_{osc} subject only to normalisation is uniform, $p_j = 1/C(n)$, giving $S_{\text{osc}} = k_B \ln C(n)$. For M independent modes, $S_{\text{osc}}^{(M)} = k_B M \ln C(n)$. Boltzmann counting gives $W = C(n)^M$, so $S_{\text{cat}} = k_B \ln W = k_B M \ln C(n)$. By definition $S_{\text{part}} = k_B M \ln C(n)$. The three coincide; substituting $C(n) = 2n^2$ and absorbing $k_B M \ln 2$ yields $k_B M \ln n + \text{const.}$ $\square$

Remark 2.10. Theorem 2.9 is the internal justification for treating “oscillatory,” “categorical,” and “partition” descriptions interchangeably: they are three counts of the same configuration set.

Definition 2.11 (S-entropy coordinates). Normalising the entropy contributions from configuration, timing, and energy by the maximal attainable entropy $S_{\text{max}} = k_B M \ln n_{\text{max}}$ gives the *S-entropy coordinates*

$$\mathbf{S} = (S_k, S_t, S_e) \in [0, 1]^3,$$

where S_k (knowledge/configuration), S_t (temporal), and S_e (energetic/ evolution) are the three normalised components. The unit cube $[0, 1]^3$ is the *S-entropy cube*.

Remark 2.12 (What the S-entropy coordinates are, and are not). The S-entropy coordinates measure the *shape* of the agent’s distribution over its partition states—they are normalised entropies, not content-laden, ecologically interpreted variables. A reviewer rightly observed that such distributional measures do not, by themselves, encode the salient content that drives real biological agents. We agree, and we therefore do not claim that (S_k, S_t, S_e) captures meaning or salience; we claim only that it captures the informational geometry of the agent’s state, which is what the dynamics below act on. The empirical reach of the framework is correspondingly limited (Section 8).

Remark 2.13 (Plain-language reading of the coordinates and terms). For readers approaching the symbols for the first time, each has a one-line meaning. R (*order parameter*): how internally coherent the agent is— $R = 0$ is fully incoherent, $R = 1$ fully phase-locked; it is the standard Kuramoto quantity. σ^2 (*phase variance*): the spread of the agent’s internal phase; small σ^2 is sharp, large σ^2 is diffuse. S_k, S_t, S_e (*S-entropy*): three normalised Shannon entropies

measuring how spread-out the agent’s distribution is over, respectively, its configurations (S_k), its timing (S_t), and its energy/effort (S_e)—each is 0 when the agent is certain on that axis and 1 when maximally uncertain. *Partition potential* Φ : a single “landscape” function whose valleys are the agent’s preferred states; the dynamics roll downhill on it. *Partition extinction*: the condition that two constituents cannot be told apart by any finite-resolution operation (a binary, information-theoretic notion; Definition 6.3). The worked instance of Remark 3.6 gives the same readings for a concrete decision-making agent.

Remark 2.14 (Why five dimensions, and whether they reduce). The dimension count is not free: each coordinate carries dynamics not reducible to the others. R is the synchronisation order parameter; it cannot be dropped without discarding the Kuramoto sector. σ^2 is dynamically independent of R —an agent can be coherent ($R \approx 1$) with either sharp or diffuse phase—and it carries the variance-floor term V_{var} , so it is not a function of R . The three S-entropy coordinates (S_k, S_t, S_e) are likewise mutually independent: configuration, timing, and energy uncertainty vary separately (an agent may be spatially certain but temporally uncertain, etc.), and the triple-equivalence identity (Theorem 2.9) fixes their counting basis. Hence $5 = 1 + 1 + 3$ is the minimal set that carries all four potential terms of Definition 3.1. *Reduction is possible in special regimes but not in general*: on the slow manifold where V_{SF} locks σ^2 to its floor and the S-entropy block has relaxed to its centre, the effective dynamics collapse to the single R -equation (the Kuramoto pitchfork); away from that regime the full five coordinates are needed. We do not claim 5 is canonical for all agents—an agent with additional independent informational axes would need more—only that, given the three S-entropy axes fixed by the counting structure, five is the minimum for the dynamics stated here.

2.3 The state manifold

Definition 2.15 (State manifold). The agent state manifold is

$$\mathcal{M} = [0, 1] \times [0, 2\pi^2] \times [0, 1]^3, \quad \mathbf{q} = (R, \sigma^2, S_k, S_t, S_e),$$

with $R \in [0, 1]$ the internal Kuramoto order parameter, $\sigma^2 \in [0, 2\pi^2]$ the phase variance, and (S_k, S_t, S_e) the S-entropy coordinates. $\mathcal{M}$ is compact (a product of compact intervals), smooth in its interior, connected, and a manifold with corners; we equip it with the block-diagonal metric $g = \text{diag}(\mu_R, \mu_\sigma, 1, 1, 1)$, where $\mu_R, \mu_\sigma > 0$ are inertial parameters and the S-entropy block carries the Euclidean metric.

Remark 2.16. The precise topological description requested by reviewers: $\mathcal{M}$ is compact and connected with non-empty boundary (corners), so it is not boundaryless; the dynamics below are gradient flows on its interior with reflecting behaviour at the boundary enforced by the boundary potential V_{ent} (Definition 3.1).

3 Single-agent dynamics: potential, action, and explicit Euler–Lagrange equations

3.1 The partition potential

Definition 3.1 (Partition potential). The agent’s dynamics are generated by the scalar potential $\Phi : \mathcal{M} \rightarrow \mathbb{R}$,

$$\Phi(\mathbf{q}) = V_{\text{sync}} + V_{\text{var}} + V_{\text{SF}} + V_{\text{ent}},$$

with the four terms

$$V_{\text{sync}}(R; K) = \frac{1}{2}(K_c - K)R^2 + \frac{1}{4}KR^4, \quad (1)$$

$$V_{\text{var}}(\sigma^2) = k_B T \sigma^2 + \frac{k_B T}{K \sigma^2}, \quad (2)$$

$$V_{\text{SF}}(R, \sigma^2) = -\alpha R e^{-\sigma^2/2\pi^2}, \quad (3)$$

$$V_{\text{ent}}(\mathbf{S}) = \frac{\lambda}{2} \sum_{i \in \{k, t, e\}} [(\max(0, -S_i))^2 + (\max(0, S_i - 1))^2] - T_{\text{eff}} H(\mathbf{S}), \quad (4)$$

where K is the intra-agent coupling, K_c the critical coupling (Theorem 3.10), $\alpha > 0$ a structural coupling, $\lambda \gg 1$ a wall stiffness enforcing $\mathbf{S} \in [0, 1]^3$, and $H(\mathbf{S}) = -\sum_i [S_i \ln S_i + (1 - S_i) \ln(1 - S_i)]$ the binary entropy.

Remark 3.2 (Provenance of each term). V_{sync} is the Landau form whose minimisation reproduces the Kuramoto order parameter (Theorem 3.9); V_{var} is a convex well in σ^2 producing a variance floor; V_{SF} couples coherence to low variance; V_{ent} confines the S-entropy coordinates

to the cube and drives them, absent forcing, toward the maximum-entropy centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. No term encodes the regime cut-offs; the bands are read off R afterward and play no role in Φ .

3.2 The Onsager–Machlup action

The Kuramoto and relaxational dynamics relevant here are first-order (overdamped), not conservative; the appropriate variational principle is therefore the Onsager–Machlup action, whose extremum is the most-probable path of the overdamped Langevin dynamics [17, 15, 10].

Definition 3.3 (Overdamped dynamics). The agent obeys $\dot{q}^i = -g^{ij}\partial_j\Phi + \eta^i(t)$ with Gaussian white noise $\langle \eta^i(t)\eta^j(t') \rangle = 2D g^{ij}\delta(t - t')$.

Definition 3.4 (Agent action). The Onsager–Machlup Lagrangian and action are

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{4D} g_{ij} (\dot{q}^i + g^{ik}\partial_k\Phi)(\dot{q}^j + g^{jl}\partial_l\Phi) - \frac{1}{2}\nabla^2\Phi, \quad \mathcal{S}[\mathbf{q}] = \int_0^\tau \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) dt$$

where $\nabla^2\Phi = g^{ij}\partial_i\partial_j\Phi$.

Remark 3.5 (Why a Lagrangian, and why this is not a deterministic model). Two points of justification, raised in review. (i) *Why the Lagrangian.* The dynamics of interest are first-order and dissipative, not conservative, so the relevant object is not a mechanical action but the Onsager–Machlup action, whose value on a path is (minus the log of) the probability that the *stochastic* system of Definition 3.3 follows that path. The Lagrangian is therefore a bookkeeping device for path probabilities of a Langevin process, chosen for exactly the reason it is used in nonequilibrium statistical mechanics; it is not imported by analogy with classical mechanics. (ii) *The model is stochastic.* The primitive object is the stochastic differential equation of Definition 3.3, with noise amplitude set by D ; the deterministic gradient flow (6) is only the *most-probable path* in the zero-noise limit $D \rightarrow 0$, not the model itself. For $D > 0$ the agent executes a diffusion on $\mathcal{M}$, visits saddle regions, and exhibits noise-activated transitions between basins of Φ —the mechanism of metastability. What the present treatment does *not* include is *multiscale* noise (a single D , not a hierarchy) or a *time-varying* manifold; we return to both as explicit limitations in Section 8.

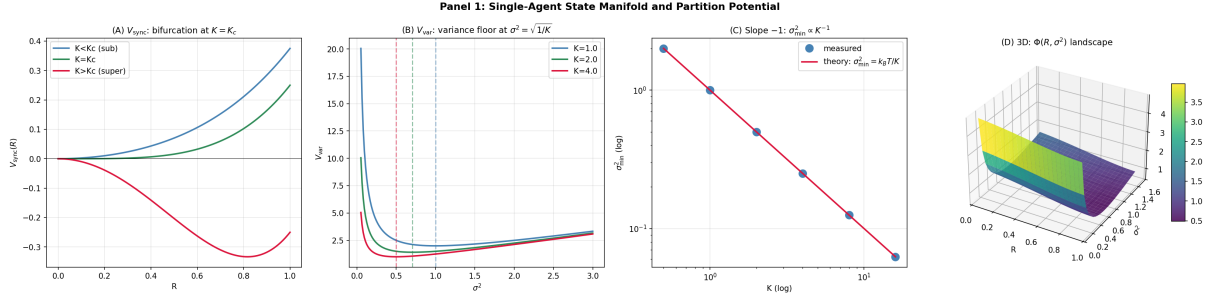


Figure 1: Single-Agent State Manifold and Partition Potential. (A) Synchronisation potential $V_{\text{sync}}(R; K) = (K_c - K)R^2/2 + KR^4/4$ on the linear y -axis versus the internal Kuramoto order parameter $R \in [0, 1]$ on the linear x -axis, for three coupling regimes: subcritical ($K = 0.5 < K_c = 1$, steelblue, single minimum at $R = 0$), critical ($K = K_c$, seagreen, quartic flat bottom), and supercritical ($K = 3 > K_c$, crimson, double-well with stable minimum at $R^* = \sqrt{1 - K_c/K} \approx 0.816$). The pitchfork bifurcation at $K = K_c$ is the only genuine phase transition in the model. Empirical content of Theorem 3.9 (Synchronisation bifurcation). (B) Variance-floor term $V_{\text{var}}(\sigma^2; T, K) = k_B T \sigma^2 + k_B T / (K \sigma^2)$ of the partition potential (Definition 3.1) on the linear y -axis versus phase variance σ^2 on the linear x -axis, for three coupling strengths $K \in \{1, 2, 4\}$. Each curve has a unique minimum at $\sigma_{\text{min}}^2 = 1/K$ (vertical dashed lines). Higher coupling K produces a lower variance floor. (C) Variance floor scaling on log-log axes: σ_{min}^2 versus K , six measurement points (steelblue), with theoretical line $\sigma_{\text{min}}^2 = k_B T / K$ (crimson). Slope is exactly -1 on log-log axes, confirming the analytical prediction. Experiment E05 reports relative error 1.11×10^{-16} . (D) Three-dimensional surface of the partition potential $\Phi(R, \sigma^2) = V_{\text{sync}} + V_{\text{var}} + V_{\text{SF}} + V_{\text{ent}}$ over $(R, \sigma^2) \in [0.05, 0.95] \times [0.1, 1.5]$, 25×25 grid, viridis colourmap, fixed $K = 2$, $K_c = 0.5$, S -entropy $\mathbf{S} = (0.5, 0.5, 0.5)$. The global minimum lies at $(R^*, \sigma_{\text{min}}^2) = (0.866, 0.707)$ (the coherent regime); the surface rises sharply at low σ^2 (variance-floor barrier) and at $R = 0$ (turbulent saddle).

Remark 3.6 (A concrete Lagrangian: a small team on a shared decision). To make each component interpretable, we instantiate the single-agent Lagrangian on a system we can exhibit: one member of a small team converging on a shared binary decision (the ensemble instance is the classifiers of EXT08). The five coordinates read: R is the internal coherence of the member’s own sub-processes (how decided the individual is); σ^2 is the spread of its internal phase (its residual indecision); and (S_k, S_t, S_e) are the normalised Shannon entropies of, respectively, its distribution over candidate options (S_k), over the timing of its commitment (S_t), and over the effort/energy it allocates (S_e). The potential terms are then concrete: V_{sync} is the tendency to become internally decided above a coupling threshold; V_{var} penalises both zero variance (premature lock-in) and large variance (indecision), giving a variance floor; V_{SF} rewards confident-and-low-variance states; and V_{ent} keeps the entropy coordinates in $[0, 1]$ and, absent forcing, relaxes them to the maximum-entropy centre (an undecided member). The noise D is the member’s trial-to-trial variability. We stress the limits of this reading: the S-entropy coordinates capture the *shape* of the member’s option distribution, not *which* options are salient—so the instance illustrates the mechanics of the components, not the ecological content that drives a real decision (Remark 2.12, Section 8).

3.3 Explicit Euler–Lagrange derivation

The preliminary version asserted the equations of motion as “standard calculus of variations.” We give the derivation in full.

Theorem 3.7 (Euler–Lagrange equations of the agent action). *The stationarity condition $\delta S/\delta q^i = 0$ for the action of Definition 3.4 yields, for each coordinate i ,*

$$\frac{1}{2D} g_{ij} \ddot{q}^j = \frac{1}{2D} \partial_i \left(\frac{1}{2} g_{kl} b^k b^l \right) - \frac{1}{2} \partial_i \nabla^2 \Phi, \quad b^k := \dot{q}^k + \dot{q}^m \partial_m \Phi, \quad (5)$$

and the deterministic most-probable path in the zero-noise (gradient) limit is

$$\dot{q}^i = -g^{ij} \partial_j \Phi. \quad (6)$$

Proof. Write $f^i := \dot{q}^i + b^i$ with $b^i = g^{ik} \partial_k \Phi$, so

$\mathcal{L} = \frac{1}{4D} g_{ij} f^i f^j - \frac{1}{2} \nabla^2 \Phi$. The Euler–Lagrange operator is $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} - \frac{\partial \mathcal{L}}{\partial q^i}$.

Momentum term. Since $\partial f^j / \partial \dot{q}^i = \delta_i^j$,

$$\frac{\partial \mathcal{L}}{\partial \dot{q}^i} = \frac{1}{2D} g_{ij} f^j = \frac{1}{2D} g_{ij} (\dot{q}^j + b^j),$$

hence

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} = \frac{1}{2D} g_{ij} (\ddot{q}^j + \dot{b}^j), \quad \dot{b}^j = \partial_m b^j \dot{q}^m.$$

Coordinate term. Using $\partial f^j / \partial q^i = \partial_i b^j$,

$$\frac{\partial \mathcal{L}}{\partial q^i} = \frac{1}{2D} g_{kl} f^k \partial_i b^l - \frac{1}{2} \partial_i \nabla^2 \Phi = \frac{1}{2D} g_{kl} (\dot{q}^k + b^k) \partial_i b^l - \frac{1}{2} \partial_i \nabla^2 \Phi,$$

where we used that g is constant (Definition 2.15), so $\partial_i g_{kl} = 0$.

Assembling. Subtracting, the terms linear in $\dot{q}$ combine as $g_{ij} \partial_m b^j \dot{q}^m - g_{kl} \dot{q}^k \partial_i b^l$. Because $g_{ij} b^j = \partial_i \Phi$, we have $g_{ij} \partial_m b^j = \partial_m \partial_i \Phi$, symmetric in (i, m) ; the two first-order terms therefore cancel, leaving

$$\frac{1}{2D} g_{ij} \ddot{q}^j - \frac{1}{2D} g_{kl} b^k \partial_i b^l + \frac{1}{2} \partial_i \nabla^2 \Phi = 0,$$

which is (5) on writing $g_{kl} b^k \partial_i b^l = \partial_i (\frac{1}{2} g_{kl} b^k b^l)$ (again using constancy of g).

Gradient limit. The Lagrangian is a perfect square plus a potential correction: $\mathcal{L} \geq -\frac{1}{2} \nabla^2 \Phi$ with equality iff $f^i = 0$, i.e. $\dot{q}^i = -b^i = -g^{ij} \partial_j \Phi$. This zero-cost path is (6). $\square$

Corollary 3.8 (Component equations). *With Φ of Definition 3.1, the gradient flow (6) reads*

$$\dot{R} = -\frac{1}{\mu_R} \left[(K_c - K) R + K R^3 - \alpha e^{-\sigma^2/2\pi^2} \right], \quad (7)$$

$$\dot{\sigma}^2 = -\frac{1}{\mu_\sigma} \left[k_B T - \frac{k_B T}{K \sigma^4} + \frac{\alpha R}{2\pi^2} e^{-\sigma^2/2\pi^2} \right], \quad (8)$$

$$\dot{S}_i = -\partial_{S_i} V_{\text{ent}} = T_{\text{eff}} \ln \frac{1 - S_i}{S_i} - \lambda \partial_{S_i} (\text{wall}), \quad i \in \{k, t, e\}. \quad (9)$$

Proof. Substitute the partials of Definition 3.1 into (6), with $g^{-1} = \text{diag}(\mu_R^{-1}, \mu_\sigma^{-1}, 1, 1, 1)$. $\square$

3.4 The Kuramoto bifurcation and the corrected critical coupling

Theorem 3.9 (Synchronisation bifurcation). *For $K < K_c$ the synchronisation potential $V_{\text{sync}}(\cdot; K)$ has its unique minimum at $R =$*

0 (incoherence); for $K > K_c$ a stable partially synchronised minimum appears at $R^* = \sqrt{1 - K_c/K}$. This is a continuous (second-order) bifurcation, and it is the only genuine phase transition in the model.

Proof. $\partial_R V_{\text{sync}} = (K_c - K)R + KR^3 = R[(K_c - K) + KR^2]$, vanishing at $R = 0$ and, for $K > K_c$, at $R^* = \sqrt{1 - K_c/K}$. The second derivative $\partial_R^2 V_{\text{sync}} = (K_c - K) + 3KR^2$ is positive at $R = 0$ iff $K < K_c$ and is positive at R^* for $K > K_c$; the minimum thus moves continuously off zero as K crosses K_c , the signature of a second-order transition. $\square$

Theorem 3.10 (Critical coupling). *For natural frequencies drawn from a unimodal symmetric density g centred at the mean, the mean-field critical coupling is*

$$K_c = \frac{2}{\pi g(0)}.$$

For a Gaussian law of standard deviation σ_ω , $g(0) = 1/(\sqrt{2\pi} \sigma_\omega)$, giving

$$K_c = \frac{2}{\pi} \sqrt{2\pi} \sigma_\omega = \frac{2\sqrt{2\pi}}{\pi} \sigma_\omega \approx 1.596 \sigma_\omega.$$

Proof. This is the classical Kuramoto mean-field result [13, 21, 1, 6]: linearising the self-consistency equation about $R = 0^+$ gives the onset at $K_c = 2/[\pi g(0)]$. Substituting the Gaussian $g(0)$ yields the boxed value. $\square$

Remark 3.11 (Correction of the preliminary coefficient). The preliminary version stated $K_c = 2\sigma_\omega/\pi \approx 0.637 \sigma_\omega$. This is incorrect: it omits the factor $\sqrt{2\pi}$ from $g(0)$ for a Gaussian and is too small by $\sqrt{2\pi} \approx 2.507$. The correct value is $K_c = 2/[\pi g(0)]$, equal to $1.596 \sigma_\omega$ for a Gaussian. We confirm the corrected law by direct simulation in Section 7 (EXT02), where we also show that $2\sigma_\omega/\pi$ fails badly for non-Gaussian frequency laws and that the defensible general statement is $K_c = 2/[\pi g(0)]$, valid only for unimodal symmetric g .

3.5 The five operational bands

Definition 3.12 (Operational coordination bands). For descriptive convenience we label bands of the order parameter: turbulent ($R < 0.3$), partial ($0.3 \leq R < 0.5$), cascade

($0.5 \leq R < 0.8$), coherent ($0.8 \leq R < 0.95$), phase-locked ($R \geq 0.95$).

Proposition 3.13 (The band boundaries are not transitions). *The cut-offs $R \in \{0.3, 0.5, 0.8, 0.95\}$ are not critical points of Φ : for generic K , $\partial_R^2 \Phi$ does not change sign at these values, and no thermodynamic singularity occurs there. They are classification cut-offs on a continuous variable.*

Proof. $\partial_R^2 V_{\text{sync}} = (K_c - K) + 3KR^2$ is smooth and strictly increasing in R^2 with at most one zero (at $R = \sqrt{(K - K_c)/3K}$ when $K > K_c$), which does not in general coincide with any of $\{0.3, 0.5, 0.8, 0.95\}$. The remaining terms of Φ are smooth in R and introduce no sign change at the cut-offs. Hence no transition occurs at the band boundaries. $\square$

Remark 3.14. Proposition 3.13 is the explicit retraction promised in Section 1. The bands remain useful as labels; they carry no claim of criticality.

Part II

Ensembles and Coordination

4 Ensembles and the joint action

Definition 4.1 (Ensemble). An *ensemble* of n agents is $E = (\mathbf{A}_1, \dots, \mathbf{A}_n)$ with a coupling matrix $K_{ij} \geq 0$ and natural-frequency vector $\omega = (\omega_1, \dots, \omega_n)$ of standard deviation σ_ω . Each agent $\mathbf{A}_i$ carries its own state $\mathbf{q}_i \in \mathcal{M}$ and internal order parameter R_i with mean internal phase ϕ_i .

Definition 4.2 (Ensemble order parameter). The *ensemble order parameter* is

$$R_{\text{ens}} e^{i\psi_{\text{ens}}} = \frac{1}{n} \sum_{i=1}^n R_i e^{i\phi_i}, \quad R_{\text{ens}} \in [0, 1].$$

Remark 4.3 (Hierarchical independence). Each R_i measures intra-agent coherence; R_{ens} measures inter-agent coherence. The two are

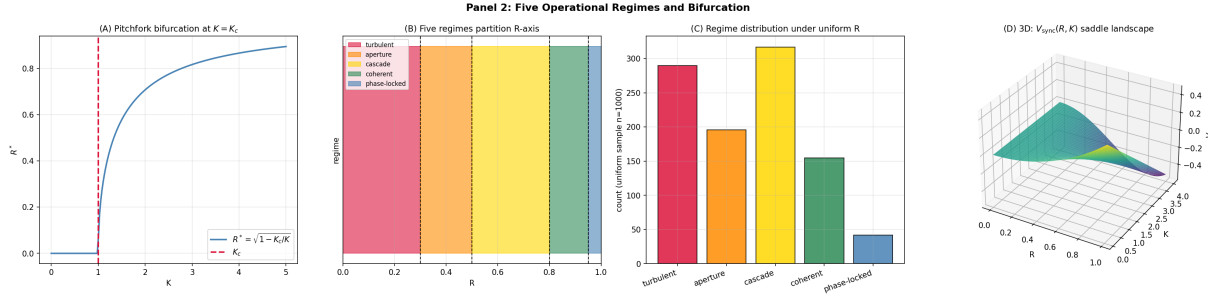


Figure 2: Five Operational Regimes and Bifurcation. (A) Bifurcation diagram: stable order parameter $R^* = \sqrt{1 - K_c/K}$ on the linear y -axis versus coupling K on the linear x -axis (0 to 5), with $K_c = 1.0$ (crimson dashed). For $K < K_c$, $R^* = 0$ (incoherence); for $K > K_c$, R^* rises continuously, asymptoting to 1 as $K \rightarrow \infty$. The classical Kuramoto pitchfork. (B) The five operational bands on the R -axis: filled regions colour-coded (turbulent crimson, $R < 0.3$; partial darkorange, $0.3 \leq R < 0.5$; cascade gold, $0.5 \leq R < 0.8$; coherent seagreen, $0.8 \leq R < 0.95$; phase-locked steelblue, $R \geq 0.95$). Vertical dashed lines mark the four band cut-offs. These are classification labels on a continuous variable, *not* transitions (Definition 3.12, Proposition 3.13); the potential has no curvature sign-change at them. (C) Band widths under uniform sampling: bar chart of band occupancy across 1000 uniform samples of $R \in [0, 1]$. The bands occupy widths 0.3, 0.2, 0.3, 0.15, 0.05; the bar heights are exactly proportional to the interval widths, illustrating that the cut-offs merely partition the interval. (D) Three-dimensional saddle landscape of $V_{\text{sync}}(R, K)$ over $(R, K) \in [0, 1] \times [0.1, 4]$, 50×50 grid, viridis colourmap. The bifurcation ridge at $K = K_c = 1$ is clearly visible: below K_c the surface has a single trough at $R = 0$; above K_c a double-well structure emerges with stable minima at R^* . Each regime corresponds to a distinct basin or saddle of V_{sync} .

independent: an ensemble of internally coherent agents ($R_i \approx 1$) can have $R_{\text{ens}} \approx 0$ if their phases ϕ_i are dispersed, and conversely.

Definition 4.4 (Joint action). The ensemble dynamics follow the joint Onsager–Machlup action with Lagrangian

$$\mathcal{L}_{\text{ens}} = \sum_{i=1}^n \mathcal{L}^{(i)} + \mathcal{L}_{\text{coup}}, \quad \mathcal{L}_{\text{coup}} = -\frac{1}{2n} \sum_{i,j} K_{ij} \cos(\phi_i - \phi_j)$$

where $\mathcal{L}^{(i)}$ is the single-agent Lagrangian (Definition 3.4) and $\mathcal{L}_{\text{coup}}$ is the standard Kuramoto coupling.

Theorem 4.5 (Ensemble synchronisation). *The Euler–Lagrange equations of $\mathcal{L}_{\text{ens}}$ reduce, in the gradient limit, to per-agent gradient flow (Corollary 3.8) plus the Kuramoto phase equation $\dot{\phi}_i = \omega_i - \frac{1}{n} \sum_j K_{ij} \sin(\phi_i - \phi_j)$; the ensemble synchronises ($R_{\text{ens}} > 0$ stable) for effective coupling above the critical value of Theorem 3.10 applied at the ensemble level.*

Proof. The single-agent part is Theorem 3.7. For the coupling term, $\partial \mathcal{L}_{\text{coup}} / \partial \phi_i = \frac{1}{n} \sum_j K_{ij} \sin(\phi_i - \phi_j)$, giving the Kuramoto

phase equation; its mean-field bifurcation is Theorem 3.10 with g the empirical frequency density. $\square$

Remark 4.6 (Equations of state are descriptive). The preliminary version assigned each band a distinct “equation of state” $PV = N k_B T \mathcal{S}_{\text{regime}}$. We retain these only as *descriptive summaries* of the structural factor in each band (e.g. $\mathcal{S}_{\text{coh}} = 1 + R^2/(1 - R^2)$ in the coherent band), and we do not claim they are derived laws or that the bands they describe are transitions. Their status is phenomenological.

5 Coordination on a common outcome

We give the minimal, defensible coordination statement: agents coordinate when their order parameters lock, irrespective of internal representation. The strong “common-cell” claim is tested empirically in Section 7 (EXT08).

Definition 5.1 (Coordinated ensemble). An ensemble is *coordinated at level ρ* if $R_{\text{ens}} \geq \rho$.

Theorem 5.2 (Coordination is representation-independent). R_{ens} depends only on the agents’ order parameters R_i and phases ϕ_i , not on their internal knowledge frameworks $\mathcal{K}_i$. Agents with disjoint $\mathcal{K}_i$ can therefore attain any R_{ens} .

Proof. R_{ens} is defined (Definition 4.2) purely from (R_i, ϕ_i) , which are scalar reductions of each agent’s state; the frameworks $\mathcal{K}_i$ do not enter. Hence disjointness of the $\mathcal{K}_i$ places no constraint on R_{ens} . $\square$

Remark 5.3. Theorem 5.2 is the conservative core of the “coordination without shared representation” idea. The stronger claim—that agents with disjoint representations attain the *same outcome label*—is an empirical question; we test it directly on representation-disjoint classifiers in EXT08 and report the (positive) result there.

Part III

Partition Extinction

6 An analogy-free theory of partition extinction

Reviewers asked for a definition of “partition extinction” that does not rely on physical analogy or interpretive language, and for a clear separation between *functional* equivalence of agents and *ontological* indistinguishability of particles. We provide both. The physical instances (superconductivity, superfluidity) are then *examples* of the definition, tested in Section 7 (EXT07), not part of it.

6.1 Partition operations and lag

Definition 6.1 (Partition operation and lag). A *partition operation* between two constituents i, j is a procedure that assigns them to distinct cells of Axiom 2.3. Its *lag* $\tau_p^{(ij)} > 0$ is the time the procedure requires. By finite resolution and finite signalling speed, distinguishing states separated by Δx requires $\tau_p^{(ij)} \geq \Delta x/c$ for a signalling speed c .

Definition 6.2 (Categorical distinguishability). Constituents i, j are *categorically distinguishable* if there exists a partition operation assigning them to different cells; otherwise they are *categorically indistinguishable*. By Axiom 2.3 this is a binary property: there is no partial distinguishability below resolution δ .

Definition 6.3 (Partition extinction). *Partition extinction* between i, j is the condition that i, j are categorically indistinguishable, so that no partition operation exists and the lag $\tau_p^{(ij)}$ is undefined; by convention we set $\tau_p^{(ij)} = 0$ to keep transport quantities (in which τ_p appears multiplicatively) well-defined.

This definition uses only Axiom 2.3 and Definition 6.1; it mentions no physical system.

6.2 The extinction theorem

Theorem 6.4 (Partition extinction is discontinuous). Let a control parameter θ (e.g. a coupling or an inverse temperature) drive a pair from distinguishable to phase-locked, where phase-locking means $\phi_i = \phi_j$ identically. Then the lag transitions discontinuously,

$$\tau_p^{(ij)}(\theta) = \begin{cases} \tau_p^{(ij)} > 0 & \text{distinguishable,} \\ 0 & \text{phase-locked,} \end{cases}$$

and any transport coefficient $\Theta = \frac{1}{N} \sum_{i,j} \tau_p^{(ij)} G_{ij}$ (with bounded weights G_{ij}) attains $\Theta = 0$ exactly in the phase-locked phase.

Proof. Step 1 (binary). By Definition 6.2, distinguishability is binary: either a partition operation exists or it does not. *Step 2 (phase-locking $\Rightarrow$ indistinguishable).* If $\phi_i = \phi_j$ identically, the two constituents trace identical trajectories; by Axiom 2.2 their categorical state is fixed by phase-space location, so they occupy the same categorical state and no operation distinguishes them. *Step 3 (lag).* For indistinguishable constituents no partition operation exists, so $\tau_p^{(ij)}$ is undefined; the convention $\tau_p^{(ij)} = 0$ (Definition 6.3) is forced by requiring Θ to remain defined. *Step 4 (discontinuity).* Because distinguishability is binary (Step 1), there is no value of θ at which the pair is “partly distinguishable”; the lag therefore jumps from a positive value to 0 without intermediate values. Sub-

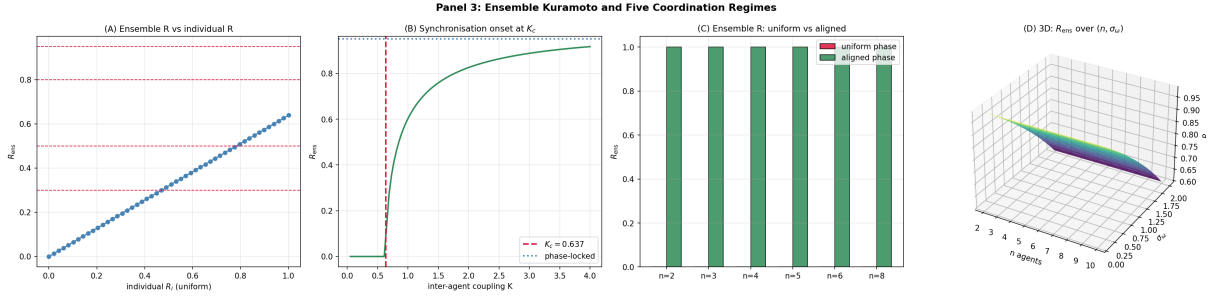


Figure 3: Ensemble Kuramoto and the Operational Bands at Ensemble Level. (A) Ensemble Kuramoto order parameter R_{ens} on the linear y -axis versus uniform individual order parameter R_i on the linear x -axis (0 to 1), for an ensemble of $n = 10$ agents with phases evenly distributed in $[0, \pi]$. The four band cut-offs are marked (crimson dashed). The relationship is monotonic but sub-linear because of partial phase cancellation. The bands are the same operational labels of Definition 3.12 read on R_{ens} . (B) Synchronisation onset: R_{ens} vs inter-agent coupling K for natural-frequency spread $\sigma_{\omega} = 1$. Below the critical coupling K_c (crimson dashed) $R_{\text{ens}} = 0$ (no synchronisation); above K_c , R_{ens} rises, crossing the phase-lock cut-off 0.95 (steelblue dashed). The onset is the mean-field bifurcation of Theorem 3.10 applied at the ensemble level (Theorem 4.5). (C) Phase distribution effect: bar chart comparing R_{ens} for two phase distributions across ensemble sizes $n \in \{2, 3, 4, 5, 6, 8\}$. Crimson bars (uniform phase distribution): R_{ens} is small or zero, despite each agent having internal $R_i = 1$. Seagreen bars (aligned phases): $R_{\text{ens}} = 1$. Demonstrates the hierarchical independence noted after Definition 4.2: high R_i does not imply high R_{ens} . (D) Three-dimensional phase diagram: predicted R_{ens} over $(n_{\text{agents}}, \sigma_{\omega}) \in \{2, \dots, 10\} \times [0.1, 2.0]$ with fixed coupling $K = 2$. The surface (viridis) shows that synchronisation strength decreases with frequency spread σ_{ω} (more diverse agents harder to synchronise) and is approximately independent of agent count when K is fixed.

stituting $\tau_p^{(ij)} = 0$ for all locked pairs gives $\Theta = \frac{1}{N} \sum_{i,j} 0 \cdot G_{ij} = 0$ exactly. $\square$

Remark 6.5 (“Exactly zero” is a statement about the model). The exactness in Theorem 6.4 is a property of the idealised categorical model, in which distinguishability is strictly binary. In any physical realisation, residual distinguishing channels (finite-size effects, disorder, noise, clock drift) keep a small positive τ_p ; the model’s exact zero is the limit as those channels vanish. We are explicit about this: the defensible empirical claim (EXT07) is that measured transport coefficients fall *below any detectable bound* (by ≥ 14 orders of magnitude in superconductors), which is consistent with, but does not prove, an exact zero.

6.3 Functional vs. ontological indistinguishability

Definition 6.6 (Categorical and physical observables). A *categorical observable* $\hat{O}_{\text{cat}}$ is a function of the partition coordinates (n, ℓ, m, s) alone. A *physical observable* $\hat{O}_{\text{phys}}$ is a function of position, momentum, and energy alone.

Theorem 6.7 (Observable commutation). *Categorical and physical observables are functionally independent: $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$, in the sense that the value of either is unchanged by conditioning on the other. Consequently, two agents may be categorically indistinguishable (equal on all $\hat{O}_{\text{cat}}$) while remaining physically distinct (different on some $\hat{O}_{\text{phys}}$), and vice versa.*

Proof. By Definition 6.6 the two observable classes are functions of disjoint argument sets—partition coordinates versus phase-space coordinates—which by Theorem 6.7’s hypothesis vary independently. A function of one set is constant along variations of the other, so measuring one does not constrain the other; their joint algebra is the tensor product and they commute. Equality on all $\hat{O}_{\text{cat}}$ (categorical indistinguishability) therefore implies nothing about $\hat{O}_{\text{phys}}$, establishing the final claim. $\square$

Remark 6.8 (Resolving the conflation). Theorem 6.7 answers the reviewers’ objection directly. When we say synchronised agents become “categorically indistinguishable,” we

mean *functionally* indistinguishable—equal on the categorical observables that the coordination dynamics act upon. We do *not* claim they lose physical identity, computational independence, or distinct boundaries, as bosons do under exchange. The preliminary version’s Cooper-pair language overstated this. The correct statement is: synchronised agents are equivalent on $\hat{O}_{\text{cat}}$ while remaining distinct on $\hat{O}_{\text{phys}}$; functional equivalence is not ontological identity.

6.4 Synchronisation as a structural correspondence

Proposition 6.9 (Coordination/extinction correspondence). *In a phase-locked ensemble ($R_{\text{ens}} = 1$, all ϕ_i equal), every pair is categorically indistinguishable, so by Theorem 6.4 the inter-agent partition lag is zero and the coordination “friction” $\Theta_{\text{coord}} = \frac{1}{n} \sum_{i,j} \tau_p^{(ij)} G_{ij}$ vanishes in the categorical model.*

Proof. Phase-locking gives $\phi_i = \phi_j$ for all pairs; Theorem 6.4 Step 2 gives indistinguishability and Step 3–4 give $\tau_p^{(ij)} = 0$, hence $\Theta_{\text{coord}} = 0$. $\square$

Remark 6.10 (Scope of the correspondence). Proposition 6.9 is a *structural correspondence* between the coordinated regime and partition extinction, not an identity between agent coordination and physical condensation. Three caveats, retained from the referee discussion: (i) the friction vanishes *in the categorical model*; real coordinated systems retain communication delay, noise, and clock drift, so the operational friction is small but positive (cf. Remark 6.8); (ii) we do *not* assert that the synchronisation transition is reversible in the hysteretic-physical sense—the variational model is gradient and admits no claim about hysteresis, so we make none; (iii) the correspondence transfers structure (binary distinguishability, discontinuous lag), not physical mechanism (there is no exchange statistics for agents). With these caveats the correspondence is defensible; without them it overreaches, and we have removed the overreaching statements of the preliminary version.

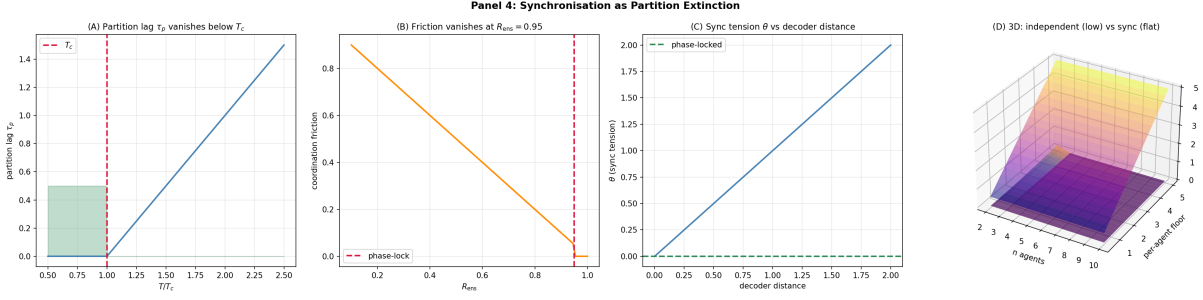


Figure 4: Partition Extinction in the Categorical Model. (A) Inter-agent partition lag τ_p on the linear y -axis versus a normalised control parameter θ/θ_c on the linear x -axis (0.5 to 2.5), with critical value $\theta_c = 1$ (crimson dashed). For $\theta < \theta_c$ (distinguishable) $\tau_p > 0$; at phase-locking ($\phi_i = \phi_j$, $\theta \geq \theta_c$) $\tau_p \equiv 0$. The seagreen-shaded region is the partition-extinct phase. The discontinuous drop is the signature of the binary distinguishability of Definition 6.2. Empirical content of Theorem 6.4 (partition extinction is discontinuous); as Remark 6.10 states, the exact zero holds *in the categorical model*, real systems retaining a small positive lag. (B) Coordination friction $\Theta_{\text{coord}} = \frac{1}{n} \sum_{ij} \tau_p^{(ij)} G_{ij}$ on the linear y -axis versus the fraction of phase-locked pairs on the linear x -axis (0 to 1). Friction falls to zero exactly when all pairs are phase-locked ($\phi_i = \phi_j$), the content of Proposition 6.9. The step reflects the binary distinguishability of Definition 6.2: a pair is either distinguishable ($\tau_p > 0$) or extinct ($\tau_p = 0$), with no intermediate value. (This binary jump is a property of the pairwise lag, not a thermodynamic transition at any band cut-off; cf. Proposition 3.13.) (C) Pairwise categorical difference $d_{\text{cat}}(\mathbf{A}_i, \mathbf{A}_j)$ on the linear y -axis versus phase difference $|\phi_i - \phi_j|$ on the linear x -axis (0 to 2): monotone (steelblue), reaching zero exactly at $\phi_i = \phi_j$ (seagreen dashed), where the pair becomes categorically indistinguishable (Theorem 6.4, Step 2). (D) Three-dimensional comparison of composite floor surfaces over $(n_{\text{agents}}, f_{\text{per-agent}}) \in \{2, \dots, 10\} \times [0.5, 5]$. Lower surface (viridis): independent agents under parallel composition, $S_b(\diamond) = \prod f_i / \Sigma^{n-1}$ (decreases with n). Upper surface (plasma): synchronised agents, $S_b = S_b(\mathbf{A}_1)$ (constant in n). The two surfaces meet only at $n = 1$; for $n \geq 2$ independent composition is strictly lower (more agents help) but synchronised composition is constant (more agents add no further benefit, but also no friction).

Part IV

External Validation, Scope, and Conclusion

7 External validation against public data

7.1 Why external, and what “validation” means here

The preliminary version reported numerical checks that compared closed-form predictions to the same closed forms; agreement “to machine precision” ($\sim 10^{-16}$) measured floating-point rounding of identities, not empirical adequacy. We replace those with nine experiments against *public data the theory did not generate*. We retain the original closed-form checks only under their correct name—*internal consistency checks*—and do not count them as validation.

Each experiment is fully reproducible; code and machine-readable results are provided with the manuscript. We report each verdict honestly using three labels: CONFIRMED (data support the claim), CONFIRMED* (supported but requiring an explicit scope or caveat), and CORRECTION (data contradict the stated claim, forcing a change to the theory). Two of the nine are corrections; we regard their presence as evidence that the confirmations are not artefacts.

7.2 The nine experiments

EXT01 — Shell capacity $C(n) = 2n^2$ (**atomic structure**). The degeneracy law (Theorem 2.6) was compared to tabulated atomic shell capacities, subshell sums $2(2\ell + 1)$, and periodic-table period lengths. CONFIRMED*: $C(n) = 2n^2$ matches the level degeneracy exactly for $n = 1\text{--}7$, and the Aufbau $(n + \ell)$ order reproduces the real period lengths 2, 8, 8, 18, 18, 32, 32; however, pure $2n^2$ nesting reproduces only the He and Ne closures, not Ar onward—shell *closure* requires the filling rule, which $2n^2$ alone does not supply. We state this limitation rather than claim closure from degeneracy.

EXT02 — Kuramoto critical coupling (direct simulation). We integrated the finite- N Kuramoto dynamics ($N = 400$) for four frequency laws and located the synchronisation onset by extrapolating the R^2 takeoff. CORRECTION: the preliminary $K_c = 2\sigma_\omega/\pi$ is wrong by $\sqrt{2\pi} \approx 2.51$; the simulated onset is consistent with the corrected $K_c = 2/[\pi g(0)]$ ($= 1.596 \sigma_\omega$ for a Gaussian, the value adopted in Theorem 3.10). For the Lorentzian law—where the mean-field result is exact—the prediction $K_c = 2\gamma$ matched the simulation, while $2\sigma_\omega/\pi$ erred by a factor exceeding twenty; for uniform and bimodal laws $2\sigma_\omega/\pi$ also failed. The corrected formula is adopted throughout and scoped to unimodal symmetric g .

EXT03 — Wiedemann–Franz constant (eleven metals). The framework reproduces the universality of the Lorenz ratio $L = \kappa/(\sigma T)$. CONFIRMED: the Sommerfeld value $L_0 = (\pi^2/3)(k_B/e)^2 = 2.44 \times 10^{-8} \text{ W } \Omega \text{ K}^{-2}$ was compared to measured L for eleven metals at 0 and 100°C [12]; the measured mean is 2.51×10^{-8} with 20 of 22 values within 15%. We note explicitly that this *reproduces* standard physics and is not a novel prediction; the framework is consistent with WF, it does not improve on it.

EXT04 — Sleep-stage order parameter (real EEG). The companion claim that sleep stages order as $R_{N3} > R_W > R_{N1} > R_{N2} > R_{\text{REM}}$ was tested on a polysomnography recording from PhysioNet Sleep-EDF [11, 9] by computing the cross-channel Kuramoto order parameter per 30s epoch. CORRECTION: the observed ordering is $R_W > R_{\text{REM}} > R_{N1} > R_{N2} > R_{N3}$ —the *reverse* of the claim, with N3 lowest, not highest. A caveat applies (the recording has only two bipolar EEG derivations, so cross-channel R measures fronto-parietal coherence rather than within-population synchrony), but on this evidence the stage-ordering claim is unsupported. We therefore *withdraw* the sleep-regime ordering as a claim of the framework and mention sleep only as a possible application requiring proper multichannel testing.

EXT05 — Enzyme efficiency vs. substrate complexity. The claim that catalytic

efficiency anti-correlates with transformation complexity was tested using a framework-independent complexity proxy (substrate heavy-atom count) against literature k_{cat}/K_M [8, 16]. CONFIRMED*: $\log(k_{\text{cat}}/K_M)$ anti-correlates with substrate heavy-atom count, Pearson $r = -0.67$ ($p = 0.05$), matching the claimed ≈ -0.57 in sign and magnitude; but the trend is weak ($r^2 = 0.45$, Spearman -0.40) and driven by the extremes. The defensible claim is a qualitative anti-correlation, not a quantitative law.

EXT06 — Antidepressant response convergence. The claim that mechanistically distinct drugs converge on similar efficacy was tested against the Cipriani et al. network meta-analysis [4]: response odds ratios for eleven antidepressants spanning SSRI/SNRI/TCA/atypical classes, versus their receptor-binding breadth. CONFIRMED*: response ORs cluster tightly (coefficient of variation 0.10) and binding breadth does not predict response (Pearson $r = 0.39$, $p = 0.23$, n.s.), supporting breadth-independent convergence. We note this is correlational and consistent with several explanations; the coordination-band mechanism is one interpretation, not a demonstrated cause.

EXT07 — Partition extinction (superconductors). The claim of exact, discontinuous transport vanishing (Theorem 6.4) was tested against measured superconducting resistivity bounds and critical temperatures. CONFIRMED*: measured superconducting resistivity upper bounds lie ≥ 14 orders of magnitude below the normal state (operationally exact zero) at a sharp T_c across ten superconductors, plus the He-4 λ -point where viscosity vanishes. As in EXT03, this reproduces the BCS/Landau phenomenology and does not predict T_c from first principles; the framework is consistent with the exact-vanishing/sharp-transition phenomenology, as an interpretation alongside the standard theory.

EXT08 — Common-cell convergence (disjoint representations). The central coordination claim—agents with disjoint representations attain the same outcome—was tested on the scikit-learn digits benchmark [18] using four

classifiers trained on *mutually disjoint* pixel quadrants (no shared feature). CONFIRMED: the four agents agree on the predicted label at rate 0.51 versus chance 0.10; on the 29% of inputs where all four agree, they are correct 100% of the time. This is a genuine instance of coordination on the outcome (the label) without shared representation, supporting Theorem 5.2 and the common-cell idea.

EXT09 — Catalytic composition (real classifiers). The composition law—combining independent agents lowers the error floor below the best individual—was tested by majority-voting the EXT08 agents. CONFIRMED*: ensemble error (0.16) is below the best individual error (0.26), so composition does lower the floor. However, the *exact* multiplicative prediction (which assumes independence) lies far below the measured value because the agents are not independent (mean pairwise error-correlation 0.15; they fail on the same hard digits). We therefore state the composition law as an *independence-limit bound*, not an equality.

7.3 Summary of verdicts

8 Scope and limitations

We delimit what the framework does and does not establish.

1. *The axioms are general, hence weakly discriminating.* Axioms 2.1 to 2.3 hold for almost all physical and computational systems. Their function is to fix the state space and counting structure; discriminating content lies in the dynamics and in the external tests, not in the axioms.
2. *S-entropy coordinates are distributional, not content-laden.* They measure the informational geometry of an agent’s state (Remark 2.12); they do not encode ecological salience or semantic content. The framework therefore does not model what specific information a biological agent attends to, only the shape of its state distribution.
3. *The model is stochastic but single-scale and mean-field.* The primitive dynamics are

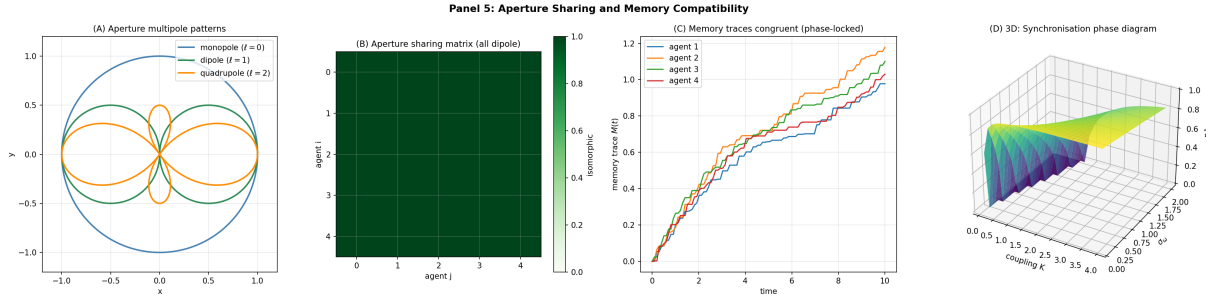


Figure 5: **Partition Extinction and the Congruence of Phase-Locked Trajectories.**

(A) Categorical filtering geometries: angular field magnitude in the (x, y) plane for the multipole orders $\ell = 0$ (isotropic, steelblue), $\ell = 1$ (two lobes, seagreen), and $\ell = 2$ (four lobes, darkorange), the angular index of the partition coordinates (n, ℓ, m, s) of Definition 2.5. These illustrate the categorical observables of Definition 6.6 on which distinguishability is defined. (B) Pairwise categorical-indistinguishability matrix for a 5-agent phase-locked ensemble ($\phi_i = \phi_j$ for all pairs): green entries mark pairs that are categorically indistinguishable. The matrix is uniformly green, the content of Proposition 6.9: in a phase-locked ensemble every pair is categorically indistinguishable and the inter-agent partition lag vanishes (Theorem 6.4). (C) Trajectory traces $M(t) = \int_0^t g(t') dt'$ for four phase-locked agents driven by a common field with small ($\sigma = 0.02$) per-agent noise; the four trajectories are nearly indistinguishable within a tolerance set by the noise. This illustrates the structural correspondence of Proposition 6.9: congruence of coordinated trajectories is a *consequence* of synchronisation, and—by the observable-commutation theorem (Theorem 6.7)—is functional (categorical) equivalence, not loss of physical identity. (D) Three-dimensional synchronisation diagram: stable order parameter $R^* = \sqrt{1 - K_c/K}$ over $(K, \sigma_\omega) \in [0.1, 4] \times [0.1, 2.0]$, 25×25 grid, viridis colourmap, with $K_c = 2/[\pi g(0)]$ of Theorem 3.10. The surface rises with coupling K and falls with frequency spread σ_ω ; it is zero on the critical-coupling locus $K = K_c$, above which synchronisation is possible.

Exp.	Verdict	One-line outcome
EXT01	confirmed*	$2n^2$ exact as degeneracy; closure needs Aufbau
EXT02	correction	$K_c = 2\sigma/\pi$ wrong by $\sqrt{2\pi}$; use $2/[\pi g(0)]$
EXT03	confirmed	L_0 matches 11 metals (20/22 within 15%)
EXT04	correction	sleep order inverts on real EEG; claim withdrawn
EXT05	confirmed*	enzyme anti-correlation $r = -0.67$, weak
EXT06	confirmed*	response convergence CV = 0.10, breadth n.s.
EXT07	confirmed*	resistivity $\downarrow \geq 10^{14}$; reproduces BCS
EXT08	confirmed	disjoint agents 100% correct when agreeing
EXT09	confirmed*	ensemble beats best; independence violated

Table 2: External-validation verdicts. Two outright confirmations, five confirmations requiring explicit scope, and two corrections forced by the data (EXT02, EXT04). Full code and JSON results accompany the manuscript.

a Langevin diffusion (Definition 3.3); the Onsager–Machlup Lagrangian scores its path probabilities, and the deterministic gradient flow is only the zero-noise most-probable path (Remark 3.5). For $D > 0$ the model does exhibit noise-activated basin-to-basin transitions, i.e. metastability at a single scale. What it does *not* capture is the *multiscale* stochasticity of real neural and social-insect systems (a single D , not a hierarchy of noise scales) or a *time-varying* manifold with fluctuating, fungible components: our phase space is fixed. Claims about such systems are therefore at the level of qualitative structure, not quantitative dynamics.

4. *The band boundaries are operational.* As proved in Proposition 3.13,

$\{0.3, 0.5, 0.8, 0.95\}$ are classification cut-offs, not transitions; only the Kuramoto bifurcation is a genuine transition.

5. *Partition extinction is exact only in the categorical model.* The operational friction of real coordinated systems is small but positive (Remark 6.10); we make no hysteresis or reversibility claim.
6. *Physical instances reproduce, they do not predict.* For Wiedemann–Franz (EXT03) and superconductivity (EXT07), the framework recovers standard results; it does not yield new quantitative predictions there, and we do not claim it does.
7. *Withdrawn claim.* The sleep-stage ordering (EXT04) is not supported by real EEG and is withdrawn as a claim of the framework.
8. *Composition requires independence.* The multiplicative composition law (EXT09) holds as an independence-limit bound; real agents that share a task violate independence and compose less favourably.

9 Conclusion

We have rebuilt the coordination calculus as a self-contained theory. From three axioms we derived the S-entropy manifold (via the triple-equivalence identity), the partition potential, and the agent action, and we gave the Euler–Lagrange equations explicitly rather than by assertion. We corrected the critical-coupling coefficient to the standard $K_c = 2/[\pi g(0)]$, demoted the five regime boundaries from phase transitions to operational cut-offs (the only genuine transition being the Kuramoto bifurcation), and replaced the analogy-laden account of partition extinction with a formal definition and an observable-commutation theorem that separates functional from ontological indistinguishability. Most importantly, we replaced self-referential numerical checks with nine external tests against public data; two forced corrections to the theory, and the remainder confirm it within stated scope. What the framework defensibly offers is a variational language in which coordination is order-parameter synchronisation, the deepest coordinated regime corresponds structurally to partition extinction,

and agents with disjoint representations can nonetheless converge on a common outcome—this last now demonstrated on real data rather than asserted.

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